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Recap

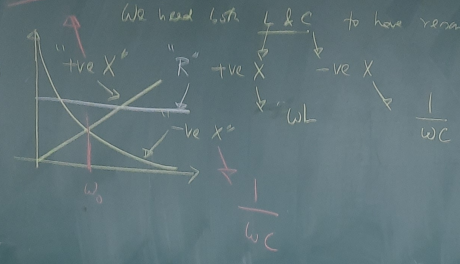
Resonance : "ω" where power delivery max

(⇒ voltage & curr in phase (⇒ "X = 0"

Resonance

Pictorially

Worked later for fun!



We need both L & C to have resonance

"no reactance"

Series RLC

$\omega_0 = \frac{1}{\sqrt{LC}}$

Half power freq. $\omega_1 < \omega_2$

↳ where $|I|$ is $\frac{1}{\sqrt{2}}$ of $|I|_{at\ res}$

(⇒ power is $\frac{1}{2}$

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$(\omega_1, \omega_2) = \omega_0\left(\sqrt{\left(\frac{1}{2Q_0}\right)^2 + 1} \mp \frac{1}{2Q_0}\right)$

$Q_0 = \frac{\omega_0}{\beta}$ ⇔ $\frac{1}{R}$ for series RLC

Quality factor

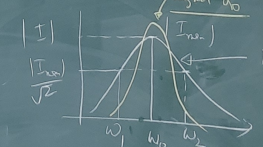
$\beta = \omega_2 - \omega_1$ is called half-power BW

$= \omega_0 \frac{1}{Q_0}$

$\omega_2 \cdot \omega_1 = \omega_0^2$ & GM is ω_0

$\frac{R}{\omega_0 L}$

$\frac{\omega_2 + \omega_1}{2} = \omega_0\sqrt{\left(\frac{1}{2Q_0}\right)^2 + 1} \geq \sqrt{\omega_2 \cdot \omega_1}$



$\left(\omega_0 > \frac{1}{2\beta}\right)$

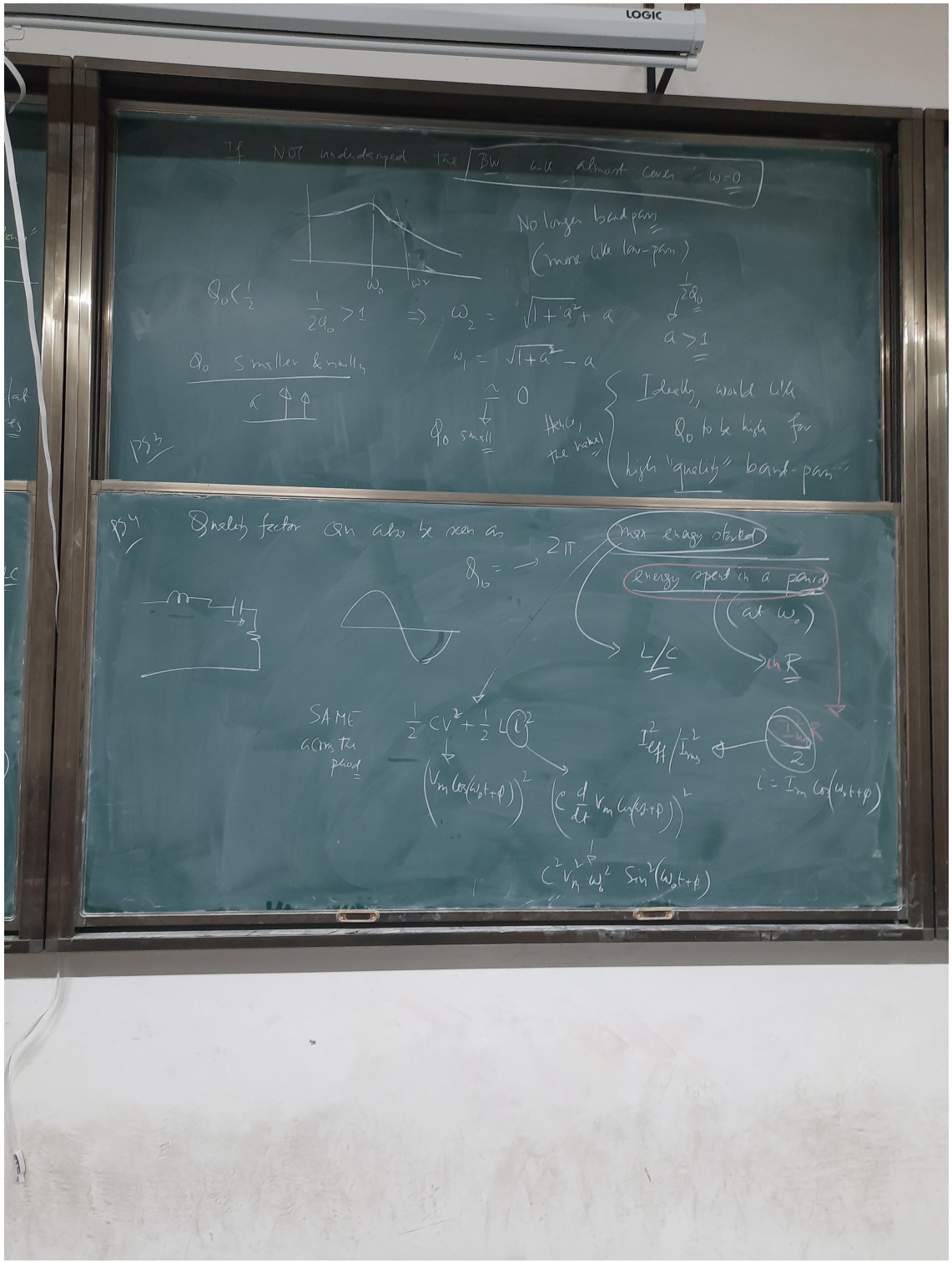
⇕

$Q_0 > \frac{1}{2}$

B gets smaller as Q_0 ↑

{ If original RLC was underdamped (⇒ $Q_0 > \frac{1}{2}$

(⇒ $\omega_2 - \omega_1 < 2\omega_0$


If NOT underdamped the BW will almost cover ω=0
No longer band pass
(more like low-pass)
$Q_0 < \frac{1}{2}$
$\frac{1}{2Q_0} > 1 \Rightarrow \omega_2 = \sqrt{1+a^2}+a$
$\frac{1}{2Q_0} = a > 1$
$\omega_1 = \sqrt{1+a^2}-a \simeq 0$
Q_0 smaller & smaller
Q_0 small
Hence, the band
Ideally, would like Q_0 to be high for high "quality" band-pass
Quality factor can also be seen as
$Q_0 = 2\pi \cdot$
max energy stored
energy spent in a period
(at ω_0)
L/C
R
SAME across the plot
$\frac{1}{2}CV^2 + \frac{1}{2}Li^2$
I_{eff}^2 / I_{rms}^2
$\frac{1}{2}$
$\left(V_m \cos(\omega_0 t+\phi)\right)^2$
$\left(C\frac{d}{dt} V_m \cos(\omega_0 t+\phi)\right)^2$
$i = I_m \cos(\omega_0 t+\phi)$
$C^2 V_m^2 \omega_0^2 \sin^2(\omega_0 t+\phi)$

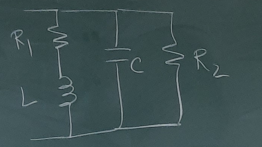
PSS

Parallel resonance
↳ band
=
ω_2/ω_1 / Half-power BW
all have the
SAME formula as series RLC
(but with a catch)

$Q_0 = \omega_0 RC$

PSS

Other resonant forms

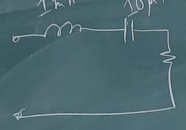


$\omega_0 = \sqrt{\frac{1}{LC} - \left(\frac{R_1}{L}\right)^2}$

(Check if $X_L = 0$)

PS6

Problem

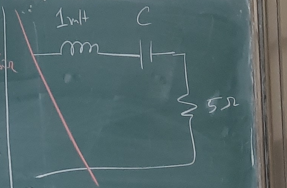


$\omega_0 = ?$
$\omega_2/\omega_1 = ?$
BW = ?

$\frac{\omega_0}{Q_0} = 10^3$ rad

$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 10 \times 10^{-6}}} = 10^6$ rad

$Q_0 = \frac{\omega_0 L}{R} = \frac{10^6 \cdot 10^{-3}}{1} = 10$



① Choose C s.t.
BW < 5K rad

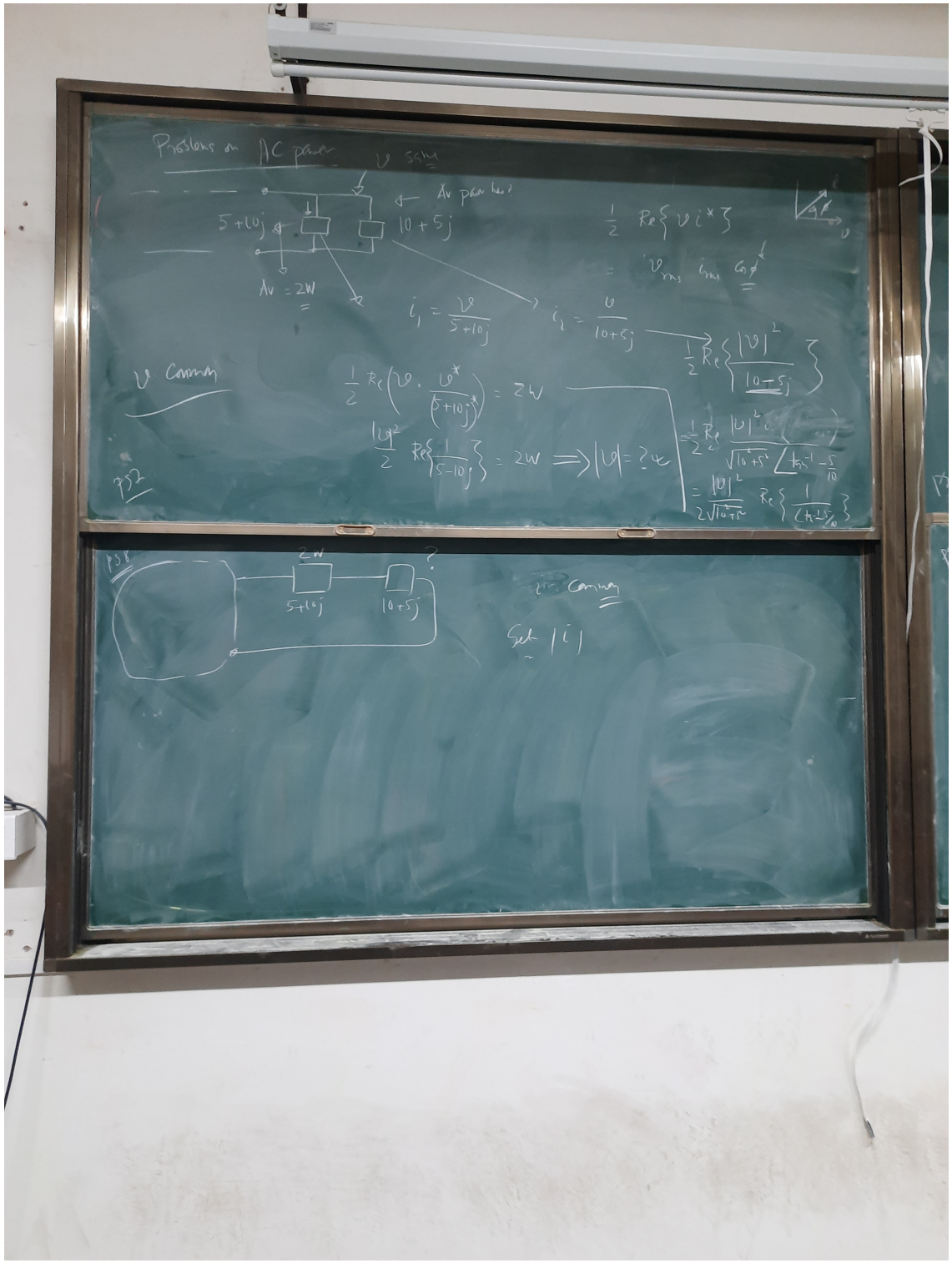
② Can you also ensure that $\omega_0 > 8$ KHz?

Load of C

Load of L

Main note:
BW / Q_0 / f depends on circuit (SC/P)

$BW = \omega_0 \frac{1}{\omega_0 RL^2} \frac{1}{RC}$


Problems on AC power
V source
Av power law?
5+10j
10+5j
Av = 2W
$i_1 = \frac{V}{5+10j}$
$i_2 = \frac{V}{10+5j}$
$\frac{1}{2} Re\{V i^*\}$
$= V_{rms} i_{rms} \cos\theta$
V Common
$\frac{1}{2} Re\left(V \cdot \frac{V^*}{(5+10j)^*}\right) = 2W$
$\frac{|V|^2}{2} Re\left\{\frac{1}{5-10j}\right\} = 2W \Rightarrow |V| = ?$
$\frac{1}{2} Re\left\{\frac{|V|^2}{10-5j}\right\}$
$= \frac{1}{2} Re\left\{\frac{|V|^2}{\sqrt{10^2+5^2}\angle \tan^{-1}\frac{-5}{10}}\right\}$
$= \frac{|V|^2}{2\sqrt{10^2+5^2}} Re\left\{\frac{1}{(10-5j)^*}\right\}$
P52
P58
2W
5+10j
10+5j
?
i: Common
Set |i|